

Subject: Mobile And Pervasive Computing (01CT0716)

Unit 1

4 Marks Questions

1. Analyze the use of WSN topologies for smart agriculture.
2. What is WLAN architecture? List its main components.
3. Compare 1G, 2G, 3G, and 4G technologies based on data rate, switching technique, and key applications
4. Compare various Wi-Fi IEEE standards with their specifications.
5. Analyze the differences between connection-oriented and connection-less services in wireless communication.
6. Design a Zigbee-based smart agriculture sensor network.
7. Differentiate between infrastructure-based and ad-hoc wireless networks with suitable examples.

6 Marks Questions

1. Evaluate Wi-Fi and Z-Wave based on their range, power consumption, and typical use cases. Which is more suitable for smart home IoT devices?
2. Apply WLAN architecture to design a wireless network layout for a university campus covering classrooms, labs, and libraries.
3. Analyze different WSN network topologies and their suitability for various applications
4. Design a wireless network layout using Wi-Fi which gives coverage around MA102 to MA115, and faculty offices. Specify access point placement and justify coverage.
5. Evaluate the suitability of Wi-Fi, and ZigBee for different IoT applications based on power, data rate, and range.
6. Compare Wi-Fi and WiMAX

8 Marks Questions

1. Compare the advantages and disadvantages of passive and active RFID tags for real time application
2. Apply the WiMAX protocol stack to the case of high-speed internet access in rural areas and explain how each layer supports service delivery
3. Evaluate the trade-offs between Wi-Fi and Z-Wave for smart healthcare systems. Consider speed, reliability, security, and energy efficiency.
4. Apply your knowledge to design a wireless network using both Zigbee and Wi-Fi in the 2.4 GHz band. How can interference be minimized?
5. Apply the Wi-Fi protocol stack to a video streaming scenario and analyze how errors or delays at different layers affect Quality of Service (QoS)
6. Discuss the differences in range, speed, and scalability between Wi-Fi and WiMAX. Provide examples of their respective use cases.
7. Describe Piconet and Scatternet in Bluetooth. Explain their structure with diagrams and role in multi-device communication.
8. Explain the architecture of a wireless sensor network. Describe key components and data flow with a relevant application.

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9. Compare RFID, NFC, and UWB technologies based on communication range, speed, power consumption, and use cases.
10. Explain the architecture of Bluetooth with a neat diagram. Describe the functions of each layer in the protocol stack.
11. Explain GSM cellular network architecture. Analyze the role of MSC and BSC in call setup and handover.

Unit 2

4 Marks Questions

1. Explain how link adaptation is applied in modern wireless networks with suitable examples.
2. What is link adaptation in wireless communication, and how does it improve network performance?
3. What is the role of random backoff in CSMA/CA, and how does it help in crowded Wi-Fi environments?
4. Identify the role of MAC layer in Ad Hoc Wireless Network and give one real life example for the same.
5. Differentiate proactive and reactive routing protocols in ad hoc networks.
6. What is Link Adaptation in Wi-Fi, and why is it needed in wireless communication?

6 Marks Questions

1. Apply the concept of medium access control (MAC) to explain how cellular networks manage multiple users accessing the same frequency spectrum without interference.

8 Marks Questions

1. Analyze the role of Adaptive Modulation and Channel Coding (AMC) in modern wireless communication systems. How does AMC improve spectral efficiency and link reliability under varying channel conditions? Support your answer with real-world examples.
2. Compare Static Routing, Distance Vector Routing (DVR), and AODV in terms of their routing strategy, adaptability, and use cases. Analyze their suitability for fixed, dynamic, and resource-constrained networks.
3. Apply LEACH protocol in a smart agriculture WSN deployment. Analyze how its clustering mechanism improves energy efficiency and data aggregation.

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Unit 3

4 Marks Questions

1. A cellular system has 80 MHz bandwidth, with 25 kHz simplex channels, and 2 MHz reserved for control. The cell area is 12 km² and the total coverage area is 4320 km². Find: Total, control, and voice channels, Control and voice channels per cell, System capacity for N=6, System capacity for N=3 and comment on the impact.
2. A mobile user is traveling at 72 km/h in a cellular network where each cell has a radius of 1.5 km. If the average call duration is 4 minutes, calculate how many handoffs the user is expected to experience during the call.
3. Analyze the difference between micro-mobility and macro-mobility protocols in terms of signaling overhead and handoff performance.
4. Apply the concept of handoff management to a high-speed train scenario. Explain which type of handoff would be most suitable and why.
5. Analyze how link layer and network layer mobility mechanisms coordinate to support seamless handoffs in mobile communication. Also, apply their roles to differentiate between macro and micro mobility scenarios.
6. A cellular system has 60 MHz bandwidth and uses 50 kHz simplex channels. With 1 MHz reserved for control, calculate: (1) total channels, (2) control and voice channels per cell (cell size: 9 km², total area: 3630 km²). Also, find the system capacity if the cluster size is reduced to 4, and comment on the impact.
7. Prove that for a hexagonal symmetry, the co-channel reuse ratio is given by $Q = \sqrt{3}N$ where $N = i^2 + ij + j^2$.

6 Marks Questions

1. How does LTE provide a stable video call to a user traveling at a very high speed (for example, 150 km/h in a train)? Explain with reference to handoff.
2. Apply the concept of mobility and handoff management in 5G to explain how a user can experience uninterrupted video streaming while traveling in a high-speed train.
3. Apply the concept of frequency reuse to design a cellular layout for a city with limited spectrum and high user density.

8 Marks Questions

1. Explain the principle of cell splitting and prove that a 12 dB transmit power reduction is required to maintain S/I when the path loss exponent $n = 4$
2. Discuss the role of VLR and HLR in supporting mobility in GSM networks. How do these databases interact during location update procedures?
3. Analyze the role of frequency reuse in wireless network systems. Illustrate with a diagram how it enhances spectrum efficiency and supports large-scale coverage in cellular communication.
4. Explain various handoff strategies used in wireless networks. Compare hard handoff and soft handoff techniques with scenarios.
5. Analyze the process of mobility management in GSM networks. What are the key differences between location update and handover procedures?
6. Why is a hexagonal cell shape preferred over square or triangular shapes in cellular network architecture? Explain with justification.

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7. Define macro mobility and micro mobility in wireless networks. Explain their differences with suitable real-life examples and scenarios in mobile IP-based systems.

Unit 4

4 Marks Questions

1. In a GSM cellular network, each frame consists of 8 time slots and the frame duration is 4.615 ms. Calculate the frame rate (frames per second). Explain with a diagram of the GSM frame structure.
2. Differentiate LTE and GSM network
3. How microcell zone concept can increase the channel capacity? Explain with the help of diagram.
4. Compare LTE and 5G
5. Explain the importance of frequency allocation in LTE and 5G cellular systems
6. Describe the GSM network architecture and explain the roles of MSC, BSC, and BTS

6 Marks Questions

1. Apply the concept of large-scale fading to analyze how shadowing and path loss influence call quality in a cellular network, and explain how the network compensates for these effects.
2. Apply the concept of small-scale fading to explain how multipath propagation affects the quality of a cellular communication link.
3. A GSM-900 operator has a 25 MHz paired band (UL: 890–915 MHz, DL: 935–960 MHz). With 0.2 MHz guard at each edge and 200 kHz channel spacing: (a) Explain ARFCN and the need for guard bands. (b) Find the number of carriers per link. (c) If two operators share the band equally, how many carriers does each get?
4. An LTE system has 10 MHz bandwidth. Each RB = 180 kHz, carrying 150 kbps. A cell has 25 users, each needing 400 kbps. (a) Can the cell support all 25 users? (b) If not, how many users can be admitted with full QoS? (c) Give one practical implication.
5. An LTE system has 15 MHz bandwidth. Each RB = 180 kHz, carrying 200 kbps. A cell has 40 users, each needing 300 kbps. (a) Can the cell support all 40 users? (b) If yes, how many RBs remain free? (c) Give one practical implication.
6. An LTE system has 20 MHz bandwidth. Each RB = 180 kHz, carrying 200 kbps. A cell has 50 users, each needing 400 kbps. (a) Can the cell support all 50 users? (b) If yes, how many RBs remain free? (c) Justify with one practical implication.
Evaluate the challenges of implementing 5G networks in India, particularly in terms of spectrum allocation and mobility management.
7. Analyze the key features of 5G networks that differentiate them from LTE, with reference to architecture and performance.
8. Explain the LTE protocol stack in detail. How is it different from GSM architecture?

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8 Marks Questions

1. Apply the cellular system architecture to explain the steps of a mobile-originated call setup, clearly stating the functions of BTS, BSC, MSC.
2. Apply your understanding of GSM and CDMA to evaluate which technology is better suited for high-density urban environments, considering capacity, handoff management, and bandwidth efficiency.
3. Design and draw the Cellular network for $i=2$, $j=2$, and $M=3$. Also highlight the cells that use same frequency band.
4. Discuss the key features and challenges of 6G wireless communication. Compare all the mobile phone generations.
5. Discuss in detail about all types of fading.
6. What is frequency reuse? How frequency reuse can use the bandwidth effectively? Analyze your answer with mathematical equations.

Unit 5

4 Marks Questions

1. How does biometric authentication improve the security of human-machine interfaces in pervasive systems?
2. Describe two examples of human-machine interfaces used in pervasive systems.
3. Define pervasive computing and list its key characteristics
4. Compare sensors and actuators in terms of their role in pervasive environments.
5. Apply the concept of context-awareness to explain why it is important in the functioning of pervasive (ubiquitous) computing devices, with suitable examples.
6. Differentiate between traditional computing and pervasive computing with suitable examples.

6 Marks Questions

1. Apply the concept of pervasive computing with cellular networks to explain how personalized healthcare services can be delivered in real time.
2. In pervasive computing environments, how can Wi-Fi and Zigbee be designed together to balance high-speed data needs and low-power IoT communication?
3. How are pervasive devices being used in modern agriculture for smart farming?
4. Apply the concept of pervasive computing to explain how smart sensors support context-aware decision making in real-world applications, with suitable examples.
5. Apply the concept of pervasive computing to explain its role and importance in modern healthcare systems, with suitable examples.

8 Marks Questions

1. Analyze the role of pervasive computing in the manufacturing industry. How does it improve efficiency, safety, and decision-making compared to traditional systems?
2. Design a way to use Wi-Fi hotspot and Bluetooth tethering together for disaster relief communication.

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3. Describe the process of setting up a Wi-Fi hotspot on a smartphone. How can this feature be applied in real-life scenarios such as online classes or fieldwork?
4. Analyze in detail the integration of biometric systems into human-machine interfaces in pervasive environments. Analyse the trade-offs between security, convenience, and privacy.
5. Discuss the ethical implications of pervasive computing in agriculture, with a focus on data collection, privacy, and the use of automated decision-making. What are the possible social and economic impacts of widespread adoption?
6. Analyse the application of pervasive computing in healthcare, focusing on remote patient monitoring, data collection via smart sensors, and patient-doctor interaction. Discuss both the benefits and the potential risks.

Unit 6

4 Marks Questions

1. How can a context-aware sensor network be applied in disaster management?
2. Explain SyncML and apply its role in achieving data synchronization across devices. Why is data synchronization important in mobile environments?
3. Explain the role of UPnP protocol in enabling context-aware services in mobile sensor networks.

6 Marks Questions

1. Apply how a pressure sensor in a context-aware sensor network can use a cellular network to adapt data transmission based on normal and abnormal readings.
2. Apply the concept of context-aware sensor networks in cellular systems to explain how smart traffic management can be achieved in a metropolitan city.
3. Apply context-aware mobile services to explain how they improve user experience with examples
4. Analyze the differences between SDP, Jini, and SLP protocols in the context of device discovery for context-aware sensor networks.

8 Marks Questions

1. Describe the architecture of a context-aware system using SyncML and smart sensors for data monitoring
2. Apply the concept of context-aware security to design a mobile sensor network using biometric authentication and open protocols

Unit 8

4 Marks Questions

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1. Suggest a future innovation in wearable devices that integrates AI/IoT. Justify its applications and impact.
2. Analyze the role of communication technologies (Bluetooth, Wi-Fi, NFC) in wearable devices.
3. What is the role sensors in wearable healthcare devices? Explain with suitable examples
4. Explain the major functions of a smartwatch and google glass.
5. How wearable devices help patient manage their health? Give two examples of what they monitor.

6 Marks Questions

1. Design a healthcare monitoring system using wearable sensors (e.g., ECG, glucose monitor). Explain architecture, data flow, and benefits.
2. Explain different types of wearable devices (smartwatches, fitness bands, smart glasses) with features.

8 Marks Questions

1. Apply the use of sensors such as accelerometers, heart rate monitors, and temperature sensors in wearable devices to explain how they support health and activity monitoring with examples

Unit 9

4 Marks Questions

1. A Body Area Network uses a set of 5 sensors, each transmitting at 50 kbps data rate to a smartphone hub. The smartphone then forwards the data via a 4G cellular link of 1 Mbps. (a) Calculate the total BAN data rate required. (b) Determine whether the cellular link can handle the aggregated data.
2. How can Body Area Networks (BANs) connected with 5G help in continuous healthcare monitoring in a smart city?
3. Apply the concept of 5G-enabled Body Area Networks to explain how remote patient monitoring can be achieved for rural healthcare.
4. How can a Body Area Network integrated with 5G cellular networks be applied to enable real-time monitoring of athletes during a sports event?
5. Explain the role of wireless communication technologies (Bluetooth Low Energy, ZigBee) in BAN.
6. What are the main types of sensors used in BAN?
7. List three medical applications of BAN with examples.
8. Define Body Area Networks (BAN) and state their primary purpose.

6 Marks Questions

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1. Describe the security and privacy issues in BAN and propose possible solutions.
2. Analyze the challenges of BAN related to energy efficiency and battery management.
3. Explain the architecture of a Body Area Network with its main components.

8 Marks Questions

1. Propose a BAN-based solution for athletes to monitor performance and prevent injuries.
2. Design a BAN system for continuous patient monitoring in a hospital. Explain the working flow and benefits.